GLA University, Mathura

Department of Computer Engineering and Applications

Assignment (Module II)

Subject: Introduction to Microprocessor

Subject Code: BCSC 0007

Submission deadline: April 30, 2026

Assignment 2.1: Introduction to Microprocessor

1. As a programmer, you are required to write an 8085-assembly language program that - Reads the 8-bit data from memory location 2000H, counts the number of bits that are 1 and stores the final count in memory location 3000H.

2. During debugging, a programmer observes that the processor accesses memory using two values: a segment register (e.g., CS, DS, SS, ES) and an offset address. The Code Segment (CS) contains 2500H, and the Instruction Pointer (IP) contains 1200H. Explain how the 20-bit physical memory address is generated in the 8086 microprocessor using the given segment and offset values and why segmentation is beneficial for such a system?

3. A robotics company is designing an automated assembly system using the 8086 microprocessor. The system performs multiple operations, while developing the control program, the programmer must use different addressing modes to efficiently access data from registers, memory, and immediate values. Identify and explain all the addressing modes of the 8086 microprocessor used in such a system.

4. A digital instrumentation system is designed using an 8085 microprocessor to display a cyclic count on an external output device. The system is required to display numbers from 0 to 9 sequentially, with a 1-second delay between each number so that the output is clearly visible to the user. The microprocessor operates at a clock frequency of 1 MHz, and the delay must be generated using a register pair-based loop. Once the count reaches 9, the system should automatically reset the counter and continue the sequence from 0, creating a continuous loop.

5. 8086 operates in two modes. What is the significance of the following signals in the maximum mode of the 8086 microprocessor? (a) NMI (b) TEST' (c) BHE'

6. An industrial automation system uses the 8254 Programmable Interval Timer interfaced with a microprocessor to generate precise timing signals. During system integration, the engineer must configure the 8254 to operate in different modes depending on the application

requirements. Name all the operating modes of the 8254. Draw a labelled diagram of the 8254 pin configuration and explain the function of each of the pins.

7. Why Microprocessor 8086 is faster than 8085? How pipelining is performed in 8086?

8. Write a control word to configure port A as input port in mode 0 and port B in mode 1 as output port. Also, a control word is given CW=CDH. Explain the conditions of ports of 8255A.

9. How many independent counters are there in 8254 Programmable Interval timer? How the counter selection is done?

10. Intel 8086 memory uses two banks to support 16-bit data transfers. Which control signals decide which memory bank(s) are accessed, and how do they determine the selection?

11. The 8259 Programmable Interrupt Controller (PIC) acts as a manager of interrupt signals up to 8 devices and forwards only the highest priority interrupt to the CPU. Describe each functional block of 8259 with proper diagram.

12. In a real-time application, data stored in memory needs to be sent continuously to an output device using the 8257 DMA controller. Explain how the 8257 performs DMA data transfer. Describe the sequence of operations including HOLD, HLDA, and data transfer control.

13. In an 8085-based microcomputer system, an 8255 Programmable Peripheral Interface (PPI) is used to interface DIP switches and output ports. The DIP switches are connected to Port B, and the output display devices are connected to Port A and Port C. Write a program to Read the data from Port B and display it on Port A. Read the data from the lower half of Port C (CL) and display it on the upper half of Port C (CU).

14. Write an assembly language program for 8085 to find the largest element in a block of 8-bit unsigned binary data. The block begins from A002H and the length of the block is stored at the memory location A001H. Also, store the smallest number at the memory location FF00H.

15. Compare and differentiate the I/O mapped I/O and memory mapped I/O.

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Assignment 2.2: Introduction to Microprocessor

Q1. You are working as a hardware design engineer in a microprocessor-based system company. Your team has decided to design a multiprocessor system using the **8086 microprocessor**, operating in **maximum mode** to support advanced features like coprocessors and bus control. Explain the **signals associated with maximum mode operation**, including their names and functions, and how they help in coordinating multiple processors and external bus controllers in the system.

Q2. You are working as a hardware design engineer in an embedded systems company. Your team is developing a microprocessor-based control system using the **8255 Programmable Peripheral Interface (PPI)** to interface input devices (like sensors) and output devices (like LEDs and motors). List and explain the **different operating modes of the 8255**, and mention where each mode can be used in the system.

Q3. You are working as a hardware engineer in a company designing a microprocessor-based embedded system using the **Intel 8086 microprocessor**. During system testing, you observe that instruction execution is faster when multiple instructions are queued. Your manager asks you to explain *the internal hardware architecture of the 8086 microprocessor with a neat diagram. Also, describe how its internal units contribute to efficient instruction execution.*

Q4. You are working as a system programmer using the **Intel 8086 microprocessor** to develop a multitasking application. While debugging, you notice that different parts of the program (code, data, and stack) are stored in separate memory locations, and the processor uses segment and offset addresses to access them. Your team lead asks you to explain the concept of memory segmentation in the 8086 microprocessor with a suitable diagram. Also, discuss the advantages of memory segmentation.

Q5 Write an assembly language program for 16 bytes of data, which are stored in memory locations at 2050H to 205FH. Transfer this entire block of data to new memory location starting at 3050H.

Q6. Explain the following signals of 8086 microprocessor with diagram:

(a) LOC$\overline{K}$
(b) TES$\overline{T}$
(c) BH$\overline{E}$
(d) S0, S1, S2
(e) QS1, QS0

Q7. Calculate the delay of given set of instructions if the microprocessor is operating with the frequency of 2 megahertz.

```
MVI B, FFH

LOOP2:  MVI C, FFH
LOOP1:  DCR C
        JNZ LOOP1
        DCR B
        JNZ LOOP2
```

Q8. Write an assembly language program to find addition with carry of 10 bytes data, which are stored in memory locations at 2050H to 2059H. Result of sum is stored in location 3000H and 3001H.

Q9. In an 8085-based microcomputer system, an 8255 Programmable Peripheral Interface (PPI) is used to interface DIP switches and output ports. The DIP switches are connected to Port B, and the output display devices are connected to Port A and Port C. Write a program to Read the data from Port B and display it on Port A. Read the data from the lower half of Port C (CL) and display it on the upper half of Port C (CU).

Q10. In a real-time application, data stored in memory needs to be sent continuously to an output device using the 8257 DMA controller. Explain how the 8257 performs DMA data transfer. Describe the sequence of operations including HOLD, HLDA, and data transfer control.

Q11. The 8259 Programmable Interrupt Controller (PIC) acts as a manager of interrupt signals up to 8 devices and forwards only the highest priority interrupt to the CPU. Describe each functional block of 8259 with proper diagram.

Q12. In a **computer system used for real-time applications (like traffic control or data acquisition)**, the CPU needs help to perform timing operations such as generating delays or counting external events. To reduce CPU workload, a programmable timer **Intel 8254 Programmable Interval Timer** is used. Draw the block diagram of 8254 and explain its working in brief.

Q13. . Write an assembly language program to find largest number of 05 bytes data, which are stored in memory locations at 2050H to 2054H and store largest number in location 3050H.

Q14. The 8259 Programmable Interrupt Controller (PIC) acts as a manager of interrupt signals up to 8 devices and forwards only the highest priority interrupt to the CPU. Describe each functional block of 8259 with proper diagram.

Q15. How 8086 is differ from 8085? Explain the structure and function of the **flag register of the 8086 microprocessor**. Describe the different flags and their roles in program execution?

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Assignment 2.3 Introduction to Microprocessor

Q.1. Describe the architecture of the 8086 microprocessors, including its key components and their functions. Additionally, explain the role of the instruction pointer, flag register, and various addressing modes in the execution of instructions on the 8086.

Q.2. Imagine you are designing a new computer system. Discuss why incorporating an interrupt controller is essential for efficient operation. Describe the functionality and features of the 8259-interrupt controller commonly used in early computer architectures.

Q.3. Identify the Mode 0 control word to configure port A and port CU as output ports and port B and port CL as input ports.

Q.4. Two machine codes 3EH and 32H are stored in 3000H and 3001H respectively. 3EH represents the opcode to load a data byte in accumulator and 32H represents the data byte to be loaded in the accumulator. Calculate the time required to execute the Opcode fetch and Memory Read Cycles and entire instruction cycle if the clock frequency is 3 MHz.

Q.5. As a hardware design engineer evaluating different timer/counter chips for an embedded system project, delve into the 8254 programmable interval timers (PIT). Discuss its significance, various operating modes, and elaborate on its areas of applications in real-world scenarios.

Q.6. Illustrate the pin diagram of the 8086 microprocessor and explain the significance of each pin in the context of memory segmentation. Discuss how memory segmentation divides the memory space into even and odd banks.

Q.7. Write a program to count from 0 to 20 H with a delay of 100 ms between each count. After the count 20 H, the counter should reset itself and repeat the sequence. Use register pair DE as a delay register. Draw a flow chart and show your calculation to set up the 100 ms delay.

Q.8. Given an 8085 with 5 MHz clock, write down a program that will provide a time delay of 1 second.

Q.9. You are a junior engineer working on a real-time control system project. Your task is to

implement a loop in assembly language on an 8085 microprocessor to achieve a precise delay of 100 μs.

```
        MVI   B, COUNT
Loop    NOP
        NOP
        DCR   B
        JNZ   Loop
```

Calculate the value of COUNT needed to achieve a loop delay of 100 μs, considering a clock frequency of 2 MHz. Express the value of COUNT in hexadecimal notation.

Q.10. Explain Intel 8255 PPI showing the different ports. Also mention the different possible modes of operating it.

Q.11. Write down a program that will store first 100 natural numbers in consecutive memory locations and store their sum after the last number.

Q.12. The 8259 Programmable Interrupt Controller (PIC) acts as a manager of interrupt signals upto 8 devices and forwards only the highest priority interrupt to the CPU. Describe each functional block of 8259 with proper diagram.

Q.13. Write a program to read the DIP switches and display the reading from port B at port A and from port CL at port CU

Q.14. Differentiate between 8085 and 8086 in terms of blocks in their architecture and highlight the key improvisation in 8086.

Q.15. Explain the following pins of 8086 pin diagram.

1. BHE
2. S0,S1 and S2
3. READY
4. MN/MX'
5. NMI